3 **(a)** Fig. 3.1 shows water in a river moving parallel to the river bank at 4.0 m / s and a canoe travelling in the river.

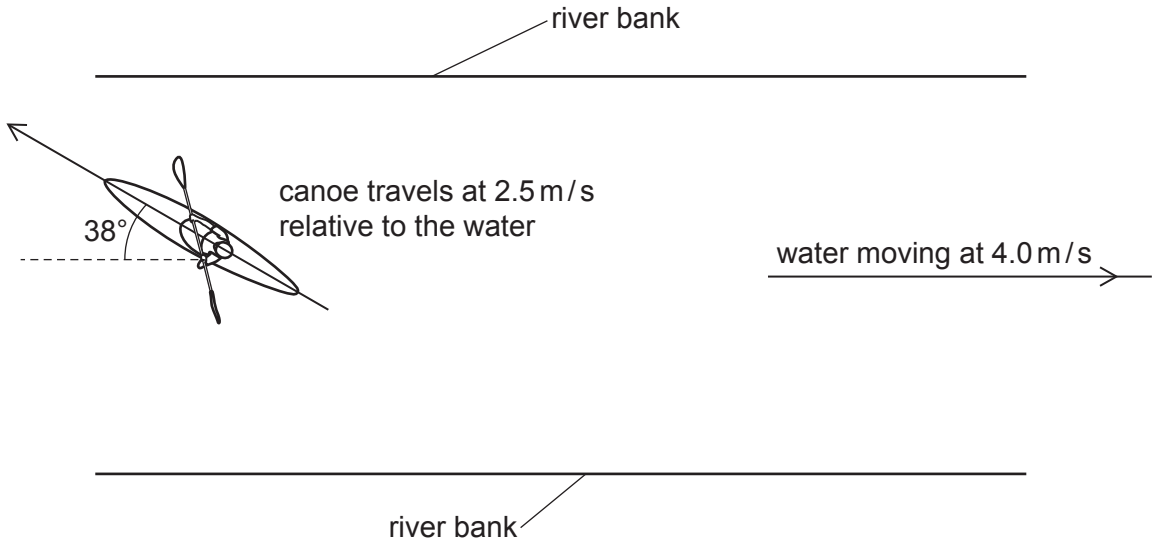


Fig. 3.1

The canoe travels at 2.5 m / s relative to the water and heads at an angle of 38° to the river bank.

Draw a scale diagram to determine the canoe's resultant velocity and state the scale you used.

scale .....

magnitude of resultant velocity .....

direction of resultant velocity (angle from the river bank) .....

[4]

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(b) The mass of the canoeist is 65 kg.

Calculate her kinetic energy when travelling on still water at 2.5 m / s.

energy = ... [2]

[Total: 6]